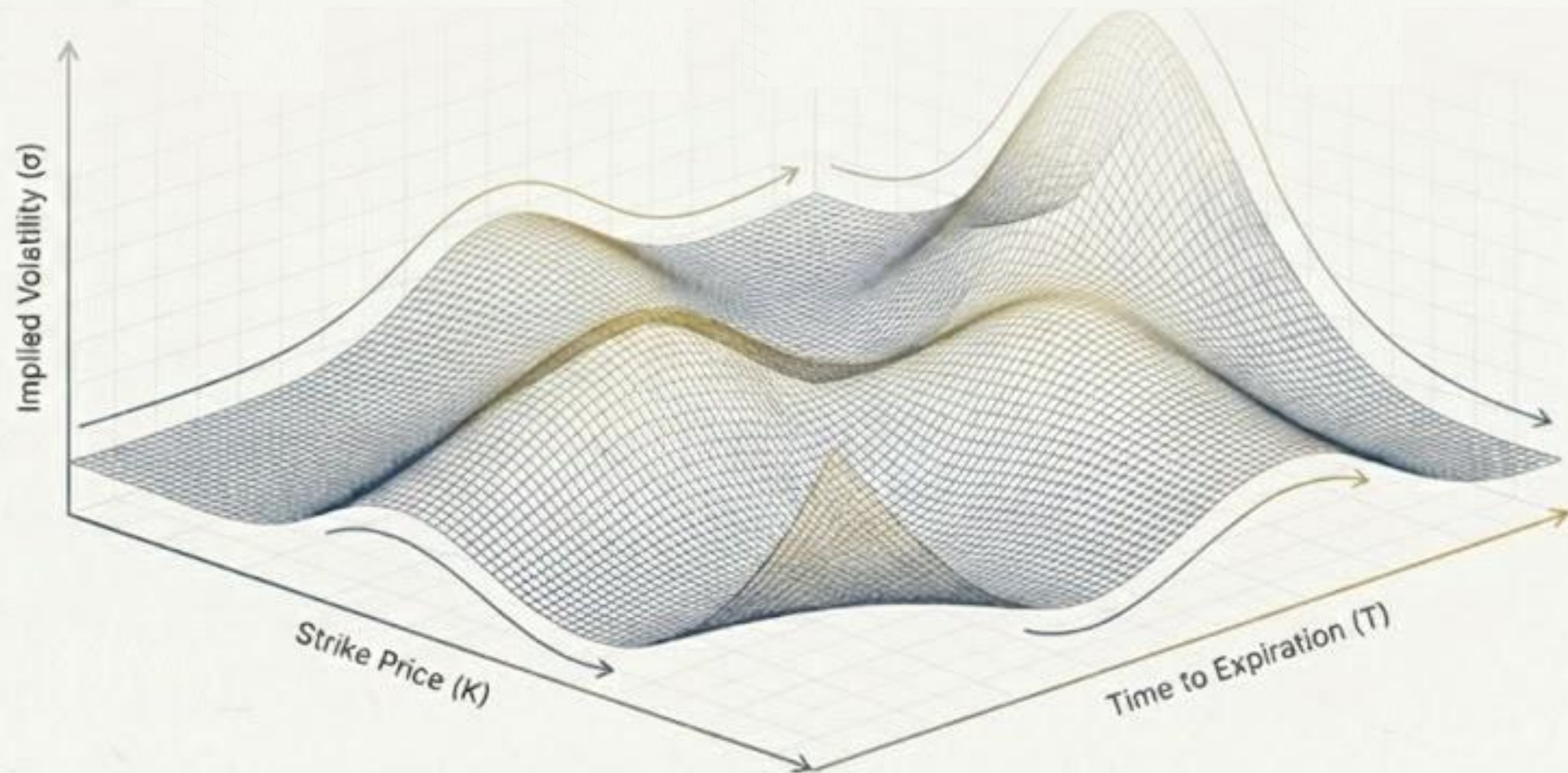


The Quest for a Better Price: A Journey Through Option Pricing Models

By Michael Fernandes



A Strategist's Guide to Option Pricing Models

The Spectrum of Option Pricing Models



The Foundation: Black-Scholes Model (1973)

The cornerstone of pricing, offering instant calculations but relying on simplified market assumptions.



Discrete Solvers: Binomial & Trinomial Trees

These models map out future price movements step-by-step, making them ideal for American options.



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Simulation Methods: Monte Carlo

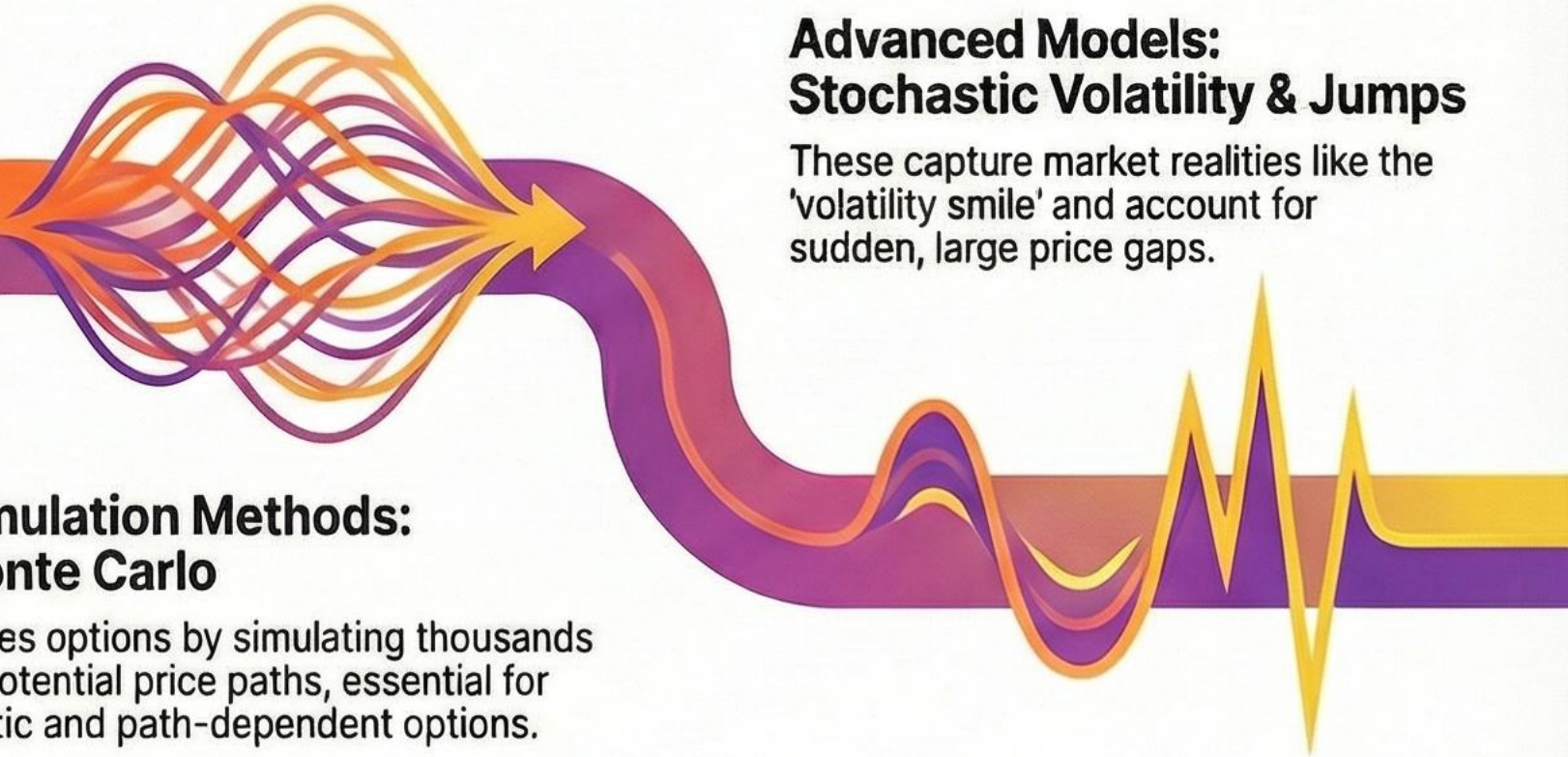
Prices options by simulating thousands of potential price paths, essential for exotic and path-dependent options.

Choosing the Right Model

Requirement	Recommended Model(s)
 Speed & Simplicity (Standard Hedging)	Black-Scholes
 American Options (Early Exercise)	Binomial Tree / Finite Difference
 Exotic or Path-Dependent Options	Monte Carlo Simulation
 Capturing the "Volatility Smile"	Heston, SABR, Local Volatility
 Pricing for Sudden Shocks (Gap Risk)	Merton Jump-Diffusion

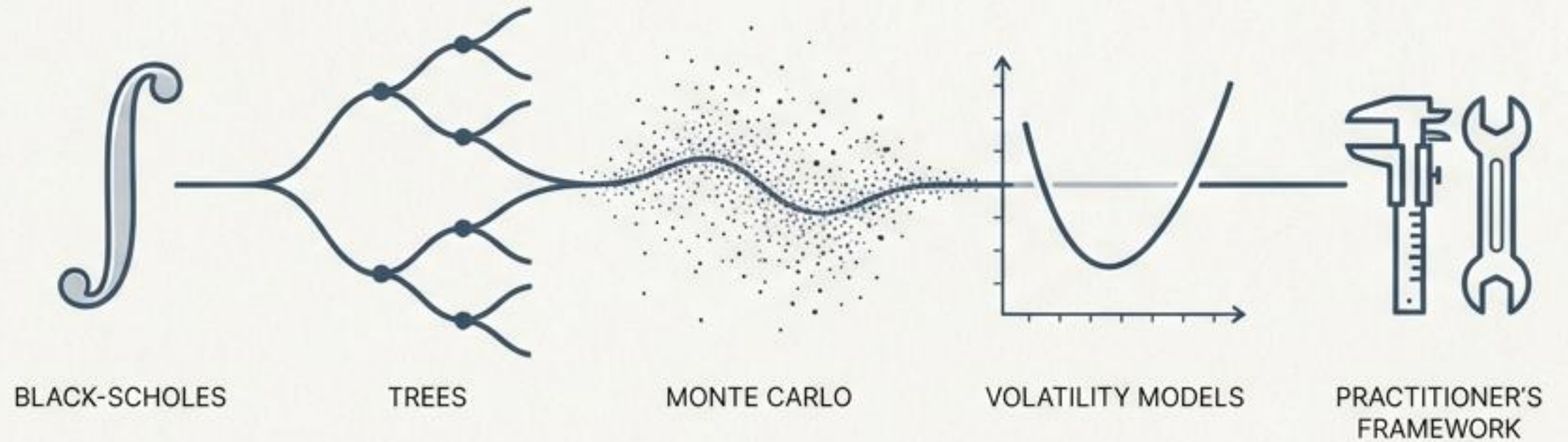
Advanced Models: Stochastic Volatility & Jumps

These capture market realities like the 'volatility smile' and account for sudden, large price gaps.



The Core Challenge: Finding the Theoretical Fair Value of an Option

Option pricing models are mathematical frameworks for determining the fair value of derivative contracts before they expire. This is a journey from elegant, closed-form solutions to sophisticated numerical methods designed to capture real-world market complexities. Our path will follow the evolution of financial thought, where each new model was developed to solve a specific limitation of its predecessor.



The Foundation: Black-Scholes and the Revolution of Risk-Neutral Pricing (1973)

Assumption: The underlying asset follows a geometric Brownian motion (GBM).

$$dS = \underbrace{\mu S dt}_{\text{Eliminated via hedging}} + \sigma S dW$$

Eliminated via hedging

Derivation: By forming a delta-hedged portfolio and applying Itô's lemma, the asset's drift (μ) is eliminated, leading to the fundamental Black-Scholes PDE:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Implication: The resulting equation demonstrates that a perfectly hedged portfolio must grow at the risk-free rate, r .

The model's groundbreaking contribution was the concept of **risk-neutral valuation**: an option's value is its expected payoff discounted at the risk-free rate, entirely independent of investor risk preferences. This was made possible by constructing a perfectly hedged portfolio.

An Elegant Solution for an Idealised World

The Closed-Form Solution (European Options)

Call Option Formula

$$C(S_t, t) = N(d_+)S_t - N(d_-)Ke^{-r(T-t)}$$

Put Option Formula

$$P = Ke^{-r(T-t)}N(-d_-) - S_tN(-d_+)$$

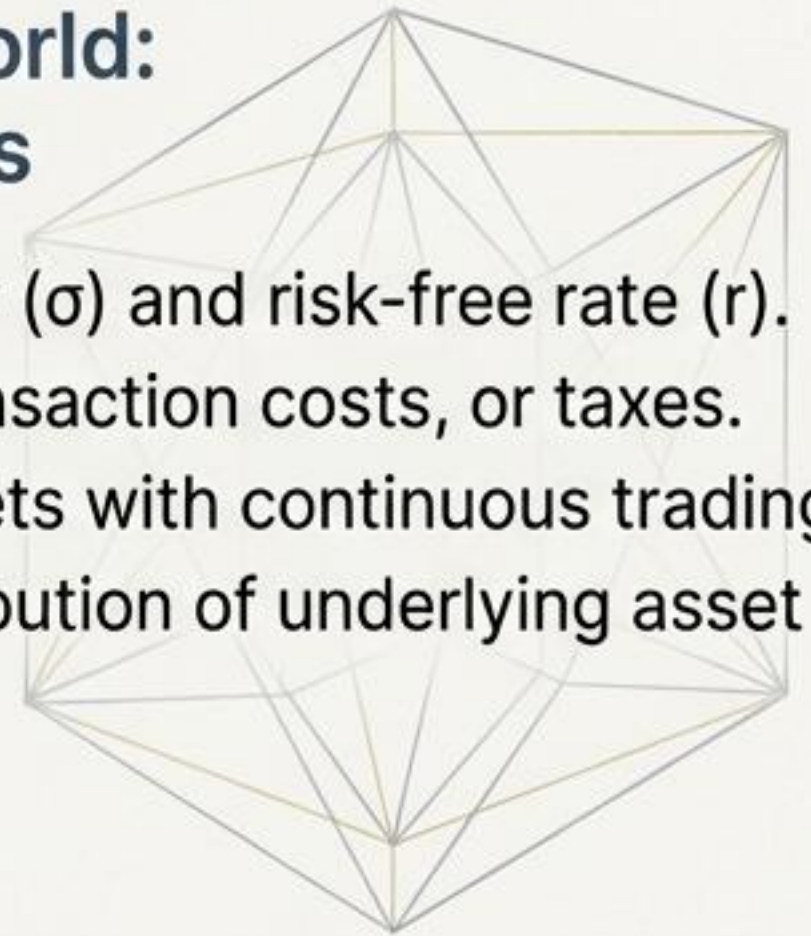
Where d_+ and d_- are defined as:

$$d_+ = \frac{[\ln(S_t/K) + (r + \sigma^2/2)(T-t)]}{\sigma\sqrt{T-t}}$$

$$d_- = d_+ - \sigma\sqrt{T-t}$$

The Idealised World: Key Assumptions

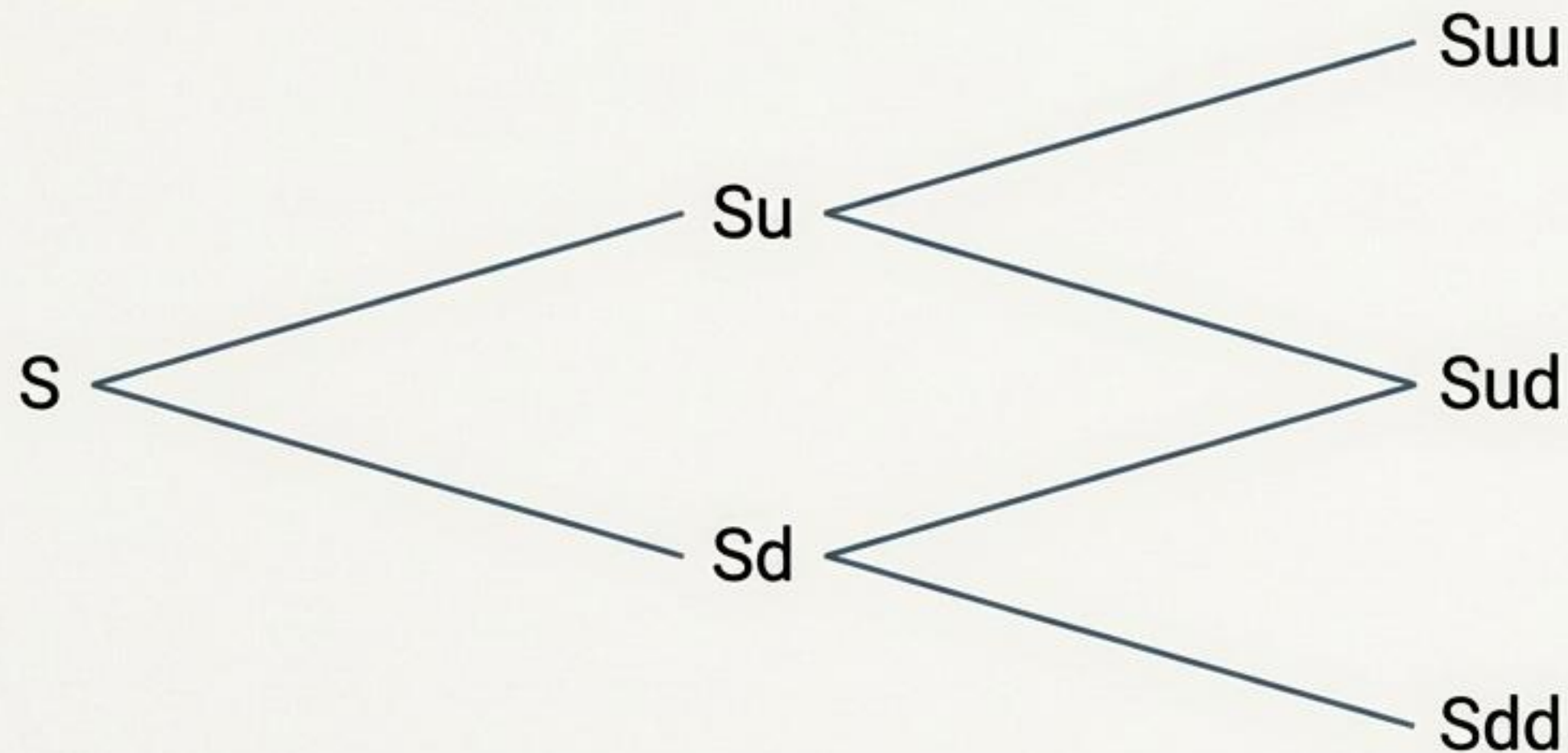
- Constant volatility (σ) and risk-free rate (r).
- No dividends, transaction costs, or taxes.
- Frictionless markets with continuous trading.
- Log-normal distribution of underlying asset prices.



Enduring Legacy

Despite its limitations, the model is the industry standard for quoting prices in terms of **implied volatility** and calculating risk sensitivities (the Greeks).

The First Challenge: Accommodating Early Exercise and Discrete Events



Problem: The Black-Scholes framework is designed for European options, which can only be exercised at expiration. It cannot value American options, where the holder has the right to exercise at any point.

Solution: The Binomial Tree model (Cox-Ross-Rubinstein, 1979) approximates the continuous GBM process in discrete steps.

At each node, the price moves up ($u = e^{\sigma\sqrt{\Delta t}}$) or down ($d = 1/u$). The option value is calculated by working backward from the payoffs at maturity, discounting at each step using a risk-neutral probability $p = (e^{r\Delta t} - d)/(u - d)$.

The Power of Trees: Flexibility and Intuitive Valuation

Key Advantage for American Options

At each node, the model compares the value of holding the option (the discounted expected future value) with the value of exercising it immediately (the intrinsic value). It selects the maximum of the two, elegantly solving the early exercise problem.

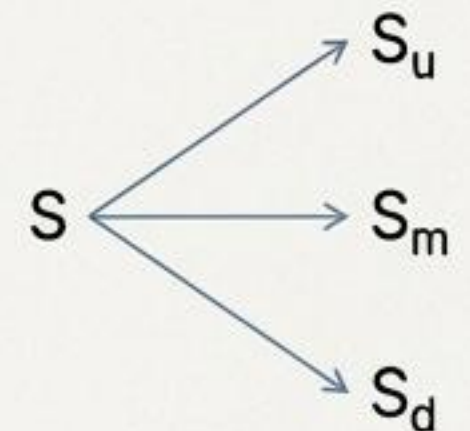
Option Value = $\max(\text{Intrinsic Value}, e^{-r\Delta t} [p \times \text{Option}_{\text{up}} + (1-p) \times \text{Option}_{\text{down}}])$

Further Advantages Over Black-Scholes

- **Dividends:** Can easily incorporate continuous or discrete dividend payments.
- **Variable Parameters:** Allows for time-varying interest rates and volatility.
- **Convergence:** As the number of time steps increases, the binomial model's price converges to the Black-Scholes price.

Extension

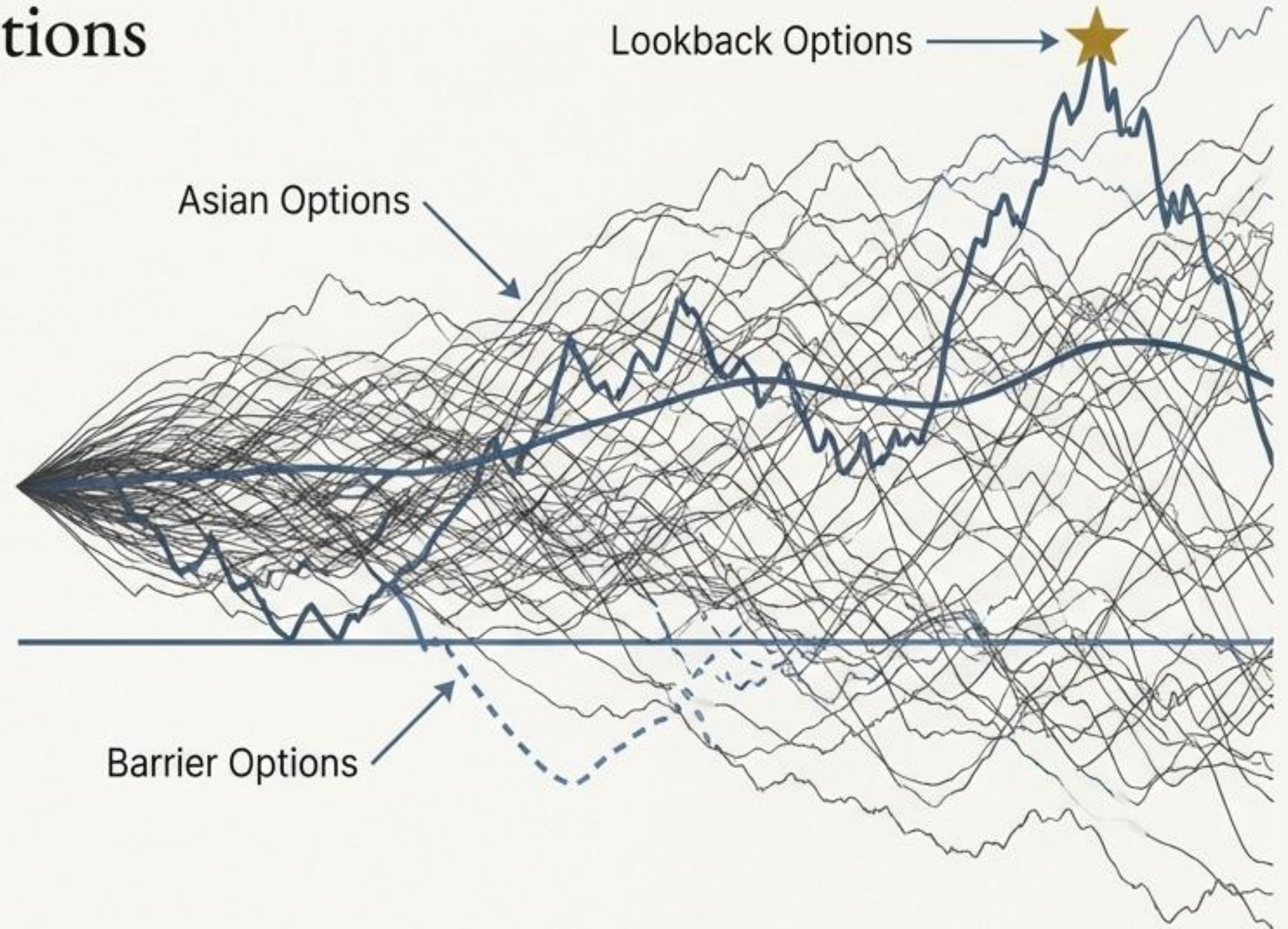
The **Trinomial Tree** adds a middle branch (price unchanged), allowing for faster convergence with fewer steps.



The Next Challenge: Pricing Path-Dependent and Exotic Options

Many options have complex payoff structures that depend on more than just the final price. Standard models cannot handle these.

- **Asian Options:** Payoff depends on the average price over a period.
- **Lookback Options:** Payoff depends on the maximum or minimum price reached.
- **Barrier Options:** Payoff is activated or extinguished if the price crosses a certain level.
- **Basket Options:** Payoff depends on a weighted average of multiple assets.



The Solution: Monte Carlo Simulation. Instead of solving a PDE, we simulate thousands of possible price paths for the underlying asset(s) to maturity.

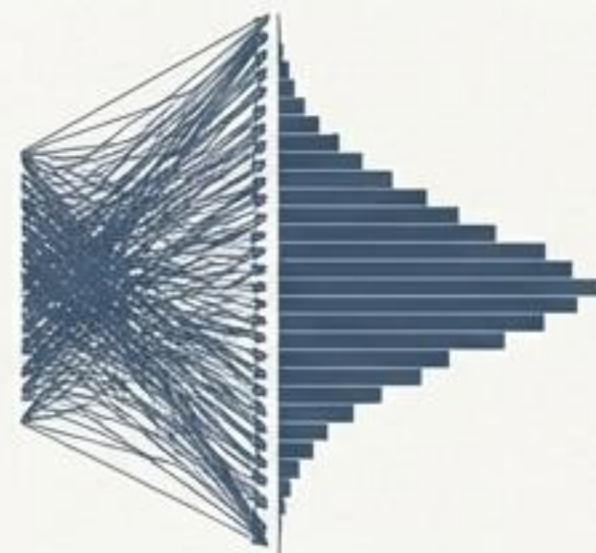
The Brute-Force Solution: Monte Carlo Simulation

1. Simulate

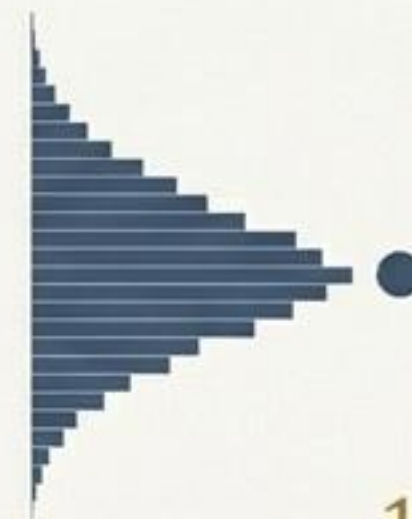


$$S(t + \Delta t) = S(t) \cdot \exp \left[\left(r - \frac{\sigma^2}{2} \right) \Delta t + \sigma \sqrt{\Delta t} \cdot Z \right]$$

2. Calculate



3. Average & Discount



$$\text{Price} = e^{-rT} \cdot \frac{1}{N} \sum \text{Payoff}_i$$

Advantages

- Handles virtually any payoff structure, including exotic and path-dependent options.
- Easily extends to multiple underlying assets.
- Avoids the “curse of dimensionality” better than grid-based methods.

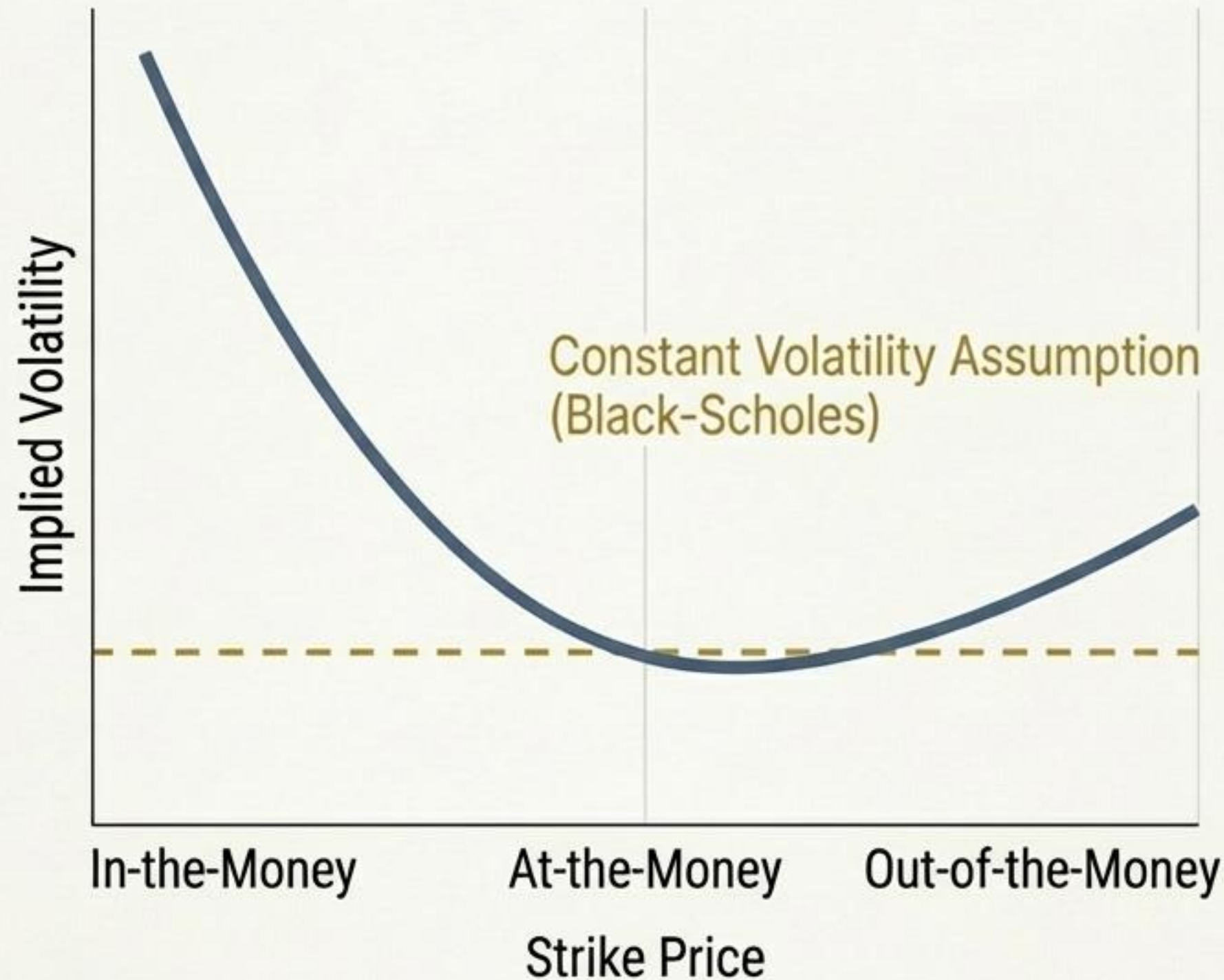
Disadvantages

- Computationally very slow.
- Results are stochastic and require confidence intervals.
- Cannot directly handle American options without advanced techniques (e.g., Longstaff-Schwartz regression).

The Elephant in the Room: The Volatility Smile

The Black-Scholes model's core assumption is that volatility is constant.

However, if we take market prices of options with the same maturity but different strike prices and calculate their implied volatility using the BS formula, we find that that it is **not** constant.



The market prices deep out-of-the-money options (representing tail events) as if they are more likely to occur than a log-normal distribution would suggest.

The Black-Scholes model systematically underprices risk.

Solution A: Letting Volatility Behave Like a Market Asset

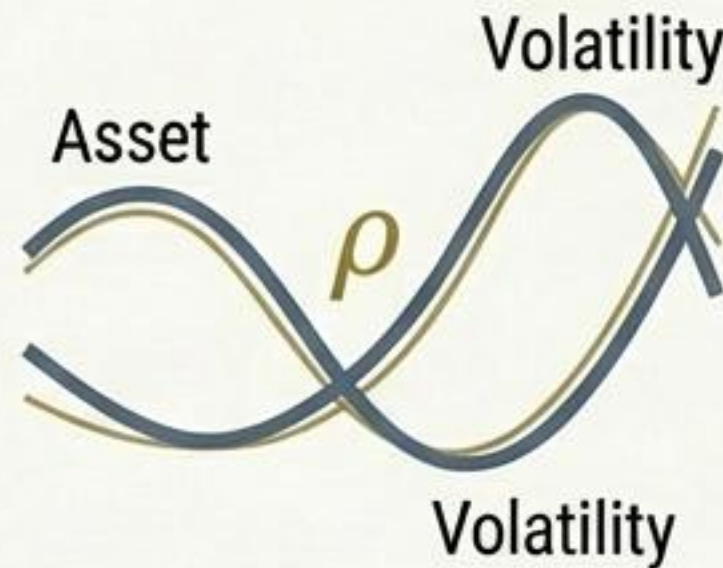
The Concept: Stochastic Volatility. These models treat volatility itself as a random variable with its own process, often correlated with the underlying asset's price.

The Heston Model

Models volatility as a mean-reverting process.

$$dS(t) = \mu S(t)dt + \sqrt{v(t)}S(t)dW(S, t)$$
$$dv(t) = \kappa(\theta - v(t))dt + \xi\sqrt{v(t)}dW(v, t)$$

- **Key Feature:** A negative correlation (ρ) between the asset and its volatility captures the **leverage effect**: volatility tends to spike when prices fall.



The SABR Model

A popular model in interest rate markets known for fast calibration.

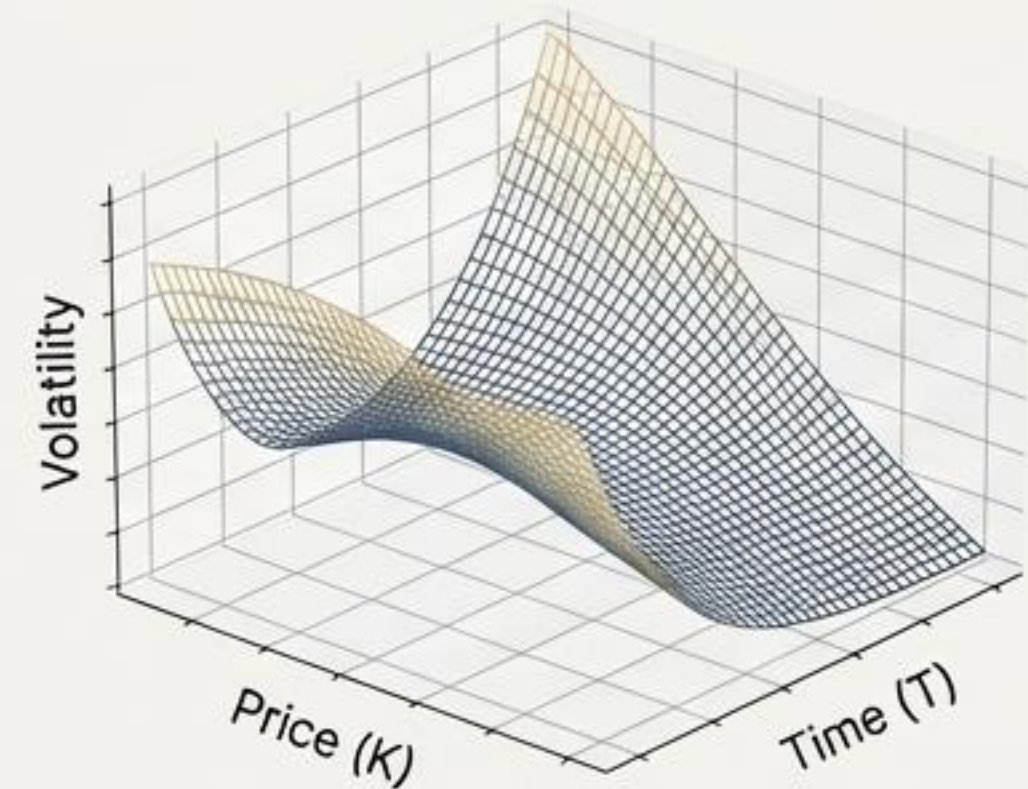
$$dF = \alpha F^\beta dW(F)$$
$$d\alpha = \nu \alpha dW(\alpha)$$

- **Key Feature:** The β parameter allows the model to interpolate between lognormal ($\beta=1$) and normal ($\beta=0$) dynamics.

Solution B: Forcing a Perfect Fit to Today's Market

The Concept: Local Volatility. This approach assumes volatility is not random, but is a deterministic function of the current asset price and time: $\sigma = \sigma(S, t)$.

$$\sigma_{\text{loc}}^2(K, T) = \frac{\frac{\partial C}{\partial T} + rK \frac{\partial C}{\partial K}}{\frac{1}{2}K^2 \frac{\partial^2 C}{\partial K^2}}$$



Advantages

- **Perfect Calibration:** By construction, it exactly reproduces the prices of all vanilla options.
- **Consistency:** Prices exotic options in a way that is consistent with the vanilla market.

Disadvantages

- **Unstable:** The derivatives in the formula amplify noise in market data.
- **Poor Dynamics:** The calibrated surface is static. It does not realistically model how the smile will move tomorrow.

Solution C: Accounting for Sudden Market Shocks

Asset prices do not always move smoothly. Events like earnings announcements or central bank decisions can cause sudden, large “jumps”.

The Merton Jump-Diffusion Model (1976)

Augments the standard GBM process with a Poisson process that triggers jumps of random size.

$$dS = \mu S dt + \sigma S dW + S(J - 1)dN$$

The model explicitly accounts for **tail risk**, explaining why out-of-the-money options are often more expensive than Black-Scholes predicts. It adds three key parameters: jump frequency (λ), mean jump size (μ_J), and jump volatility (σ_J).



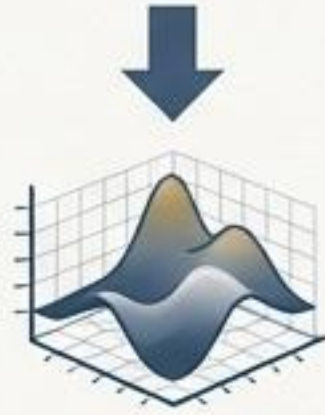
The Practitioner's Framework: Choosing the Right Tool for the Job

Requirement	Recommended Model	Reason
Speed (Greeks, hedging basis)	Black-Scholes	Instant, closed-form, market convention for implied vol.
American Options	Binomial/Trinomial Trees or Finite Difference Methods	Natively handles early exercise decisions.
Exotic & Path-Dependent Options	Monte Carlo Simulation	Handles arbitrary payoff structures and high dimensions.
Fit the Volatility Smile	Heston, SABR, or Local Volatility	Explicitly model the variation of implied vol by strike.
Interest Rate Derivatives	SABR	Standard for fast calibration and smile fitting in rates markets.
Capture Jump Risk (e.g., earnings)	Merton Jump-Diffusion	Explicitly models tail events and gap risk.
Regulatory Risk (VaR, scenarios)	Monte Carlo (with stochastic models)	Can simulate portfolio-wide risk under complex scenarios.

A Layered Reality: How Models Are Used in Practice



Market Prices: Collect bid/ask prices for liquid vanilla options.



Implied Volatility Surface: Use Black-Scholes (as a convention) to convert all prices into a surface of implied volatilities. Smooth the surface (e.g., using **SVI**).



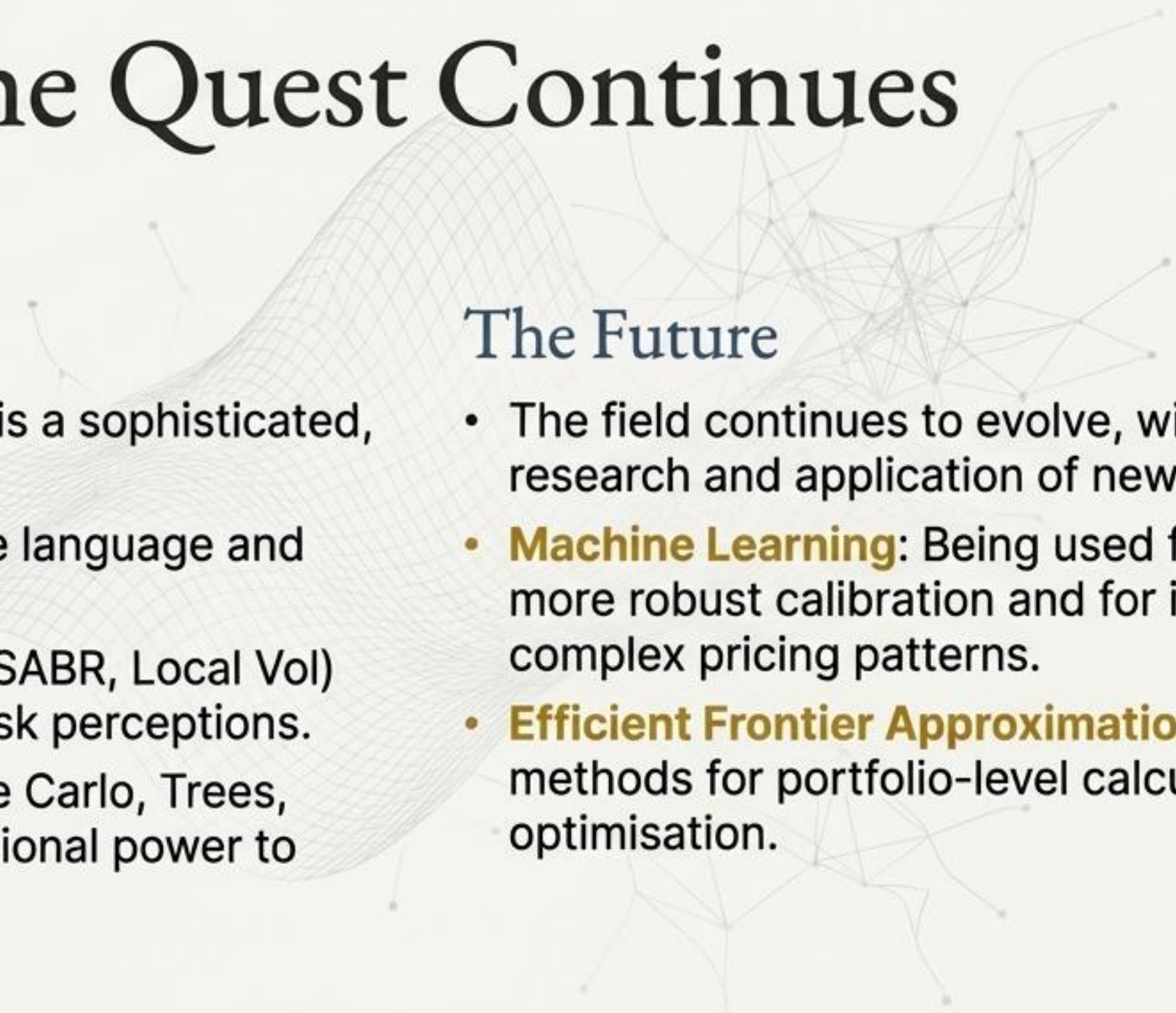
Advanced Model Calibration: Calibrate an advanced model (e.g., **Heston**, **SABR**) to this smoothed implied volatility surface.



Exotic Pricing & Risk: Use the calibrated advanced model to price complex, non-vanilla derivatives and compute their **Greeks** for hedging.

The Principle: This ensures that exotic options are priced **consistently** with the liquid, observable vanilla market, preventing arbitrage opportunities.

The Quest Continues



Summary

- Modern derivatives pricing is a sophisticated, layered approach.
- **Black-Scholes** provides the language and baseline for risk.
- **Volatility Models** (Heston, SABR, Local Vol) capture the market's true risk perceptions.
- **Numerical Methods** (Monte Carlo, Trees, FDM) provide the computational power to handle any structure.

The Future

- The field continues to evolve, with active research and application of new techniques.
- **Machine Learning**: Being used for faster, more robust calibration and for identifying complex pricing patterns.
- **Efficient Frontier Approximations**: New methods for portfolio-level calculations and optimisation.

The journey from simple formulas to complex simulations reflects the market's own evolution—an unending effort to model risk more accurately.